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Exploring innovation management systems: A systematic literature review and bibliometric analysis

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ABSTRACT

An Innovation Management System is a structured framework designed to manage innovation within organizations to achieve strategic goals. This study conducts a systematic literature review of 118 articles (2003–2023) retrieved from Scopus and the Web of Science database. Through VOSViewer Software, a bibliometric analysis was conducted, with foundational articles, keywords and authors also identified. Through a bibliographic coupling and content analysis, we identify four clusters: Theories and Conceptual Frameworks, Methodological Approaches, Research Areas & Sectors and Current and Future Impacts. Our research shows a focus on the theorical aspect of IMS, rather than practical applications. These findings and their implications are discussed.

1. Introduction

Innovation is one of the most important aspects of any forward-thinking company, serving as the driving force behind growth, competitiveness, and long-term sustainability (Banmairuroy et al., 2022). In today's rapidly evolving business landscape, organizations require to fulfill the ever-changing needs of their customers and anticipate their future demands, or else fail the risk of becoming obsolete (Muñoz et al., 2022; Ed-Dafali et al., 2023). They adapt to emerging technologies, market dynamics, and global challenges, allowing them to remain agile and resilient in uncertainty (Bogers et al., 2019). Furthermore, innovation can enhance operational efficiency, reduce costs, and lead to the development of groundbreaking products and services that capture market share (Johansen and Isaeva, 2021; Boly et al., 2022). An Innovation Management System (IMS) plays a crucial role in supporting and facilitating an organization's implementation of innovation and ensures that everyone is aligned with the organization's goals and understands their roles in the process.

An Innovation Management System (IMS) is a guiding framework that helps organizations to strengthen their innovative capabilities by providing a structured approach or a set of processes to implement and systematically manage innovation (Chiesa et al., 1996; Arno et al., 2012; Cortimiglia et al., 2015). The goal of an IMS is to create an environment that fosters creativity, encourages the development of new ideas, and effectively transforms those ideas into tangible outcomes, such as new products, services, or processes (Morel and Claire, 2021). An IMS includes all the elements and their interactions that are needed for an organization to establish the purpose of effectively and sustainably achieving innovation-driven results, while maintaining a continuous monitoring and evaluation of all processes and the performance of the implementations. Besides, it provides valuable insights for identifying areas for improvement and making informed decisions for future innovations (Mavroeidis and Tarnawska, 2017; Endres et al., 2022).

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Current research on Innovation Management Systems (IMS) focuses on several dynamic and evolving areas that reflect its critical role in fostering organizational innovation. Key aspects to consider include lack of sector-specific insights that address unique industry needs, along with insufficient examination of standards like ISO 56002 across diverse environments (Kihlander et al., 2024). Methodological limitations, such as reliance on case studies and a lack of longitudinal studies, further hinder a comprehensive understanding of IMS (Chapa et al., 2023). In previous work, (Khosravi et al., 2019) focus on the concept of Management Innovation and empirical studies, while (Idris and Durmuşoğlu, 2021) provides a study on standards of IMS; both literature reviews contribute to the subject of this paper, but this work focuses more on understanding the potential of IMS, a more refined content analysis of all contributions, the inclusion of ISO 56000 and its impact on the literature, trends, gaps, and future research. All research questions expect to fill the gaps between previous work and current work, especially the lack of inclusion of ISO 56002 as a standard in the implementation of IMS, focusing only on managerial innovation topics.

This study aims to explore key components and frameworks of IMS identified in existing literature, investigate emerging trends and challenges in the field, assess alignment or divergence between academic understanding and practical applications of innovation management systems, and identify gaps or inconsistencies in the current literature, thereby proposing potential avenues for future research. Fig. 1 shows the course of our study. The first step includes the literature review and the selection of articles selected for the research, which follows a systematic revision, as well the overview and analysis, which implicates a bibliometric and content analysis of the sources, follow by the discussion and findings of our investigation.

The paper is organized as follows. Section 2 presents a theoretical overview of the existing literature on the subject, Section 3 illustrates the research methodology and Section 4 shows the results and main findings of the research, while Section 5 discusses them and presents the research agenda. Finally, Section 6 presents the conclusions, limitations and future research.

2. Literature review

Innovation management systems (IMS) are organizational frameworks designed to foster and systematize innovation within companies. These systems typically comprise an innovation process and supporting elements (Cortimiglia et al., 2015). The ISO 56002 standard, introduced in 2019, provides a common language and framework for building innovation capabilities (Hyland and Karlsson, 2021). Key considerations when designing an IMS include understanding a company's innovation ambitions, analyzing systemic dimensions, and implementing balanced control mechanisms (Kihlander et al., 2024). IMS components vary across industries, with mining companies requiring specific internal and external innovation capabilities to adopt knowledge and develop innovation-driven strategies (Lazarenko et al., 2019). As businesses recognized the need not only to encourage creative ideas but also to manage the process of turning those ideas into practical innovations, the term "Innovation Management" gained prominence (Mir et al., 2016). While there may not be a specific origin point for the term, it has become increasingly prevalent in academic literature, business discourse, and organizational practice, but it has been widely recognized with the following terms: Innovation System, which can refer to the broader context of innovation within a region or industry, it is sometimes used synonymously with Innovation Management System, Innovation Framework, often used to describe a structured approach or model for managing innovation within an organization, and Innovation Process, which focuses on the systematic steps and stages involved in generating, developing, and implementing innovations (Idris and Durmuşoğlu, 2021).

The Innovation Management System (IMS) benefits significantly from integration with frameworks such as Open Innovation, Triple Helix Model, Resource-Based View (RBV), and Knowledge Management (KM). Open Innovation enriches IMS by incorporating both internal and external ideas, emphasizing partnerships and ecosystems to broaden innovation potential (Bogers et al., 2019). Similarly, the Triple Helix Model highlights the strategic collaboration between academia, industry, and government, aligning IMS efforts with societal and economic goals through external stakeholder engagement (Mavroeidis and Tarnawska, 2017). These approaches ensure IMS remains adaptable and ecosystem-oriented. Internally, IMS aligns with the Resource-Based View (RBV) by leveraging unique organizational resources and capabilities, such as talent and intellectual capital, to gain competitive advantage (Klius and Chizh,

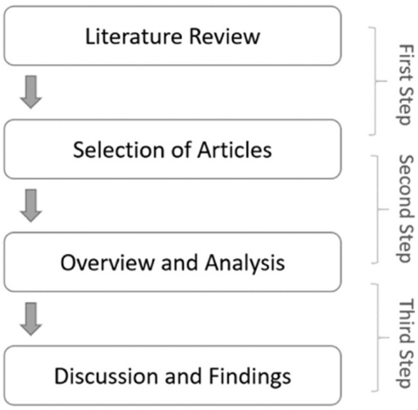

Fig. 1. Flow of the study. *Source: Author's own contribution.*

2017). Knowledge Management (KM) further complements IMS by institutionalizing knowledge flow, fostering organizational learning, and enabling data-driven innovation processes. Together, these frameworks provide IMS with a robust theoretical and practical foundation, supporting its role in systematizing innovation and driving sustainable growth (Banmairuroy et al., 2022). Total Quality Management (TQM) and Innovation Management Systems (IMS) relate to each other through their shared focus on improving organizational performance, fostering a culture of continuous improvement, and achieving strategic objectives. Both frameworks emphasize systematic processes and stakeholder involvement but differ in their core objectives and methodologies (Solaimani et al., 2019).

ISO standards are globally recognized frameworks, guidelines, or specifications established by the International Organization for Standardization (ISO). These standards are designed to ensure quality, safety and efficiency across various industries and processes. ISO standards are voluntary but widely adopted by organizations worldwide to enhance operations, meet regulatory requirements, and build customer trust (Mavroeidis and Tarnawska, 2017). ISO 56002 is a standard developed by the International Organization for Standardization (ISO) entitled "Innovation management — Innovation management system — Guidance." This standard provides guidelines and a framework for establishing, implementing, maintaining, and continually improving an innovation management system within an organization. ISO 56002 was published in 2019 and is intended to assist organizations in developing a systematic approach to managing innovation and it is highly important in IMS for several compelling reasons. First, it provides a standardized structure that offers guidance on establishing, implementing, maintaining, and continually improving innovation management systems. This international standard fosters consistency and coherence in how organizations approach innovation, promoting a systematic and holistic perspective. Additionally, ISO 56002 aligns with other ISO management system standards, enabling organizations to integrate innovation seamlessly into their overall management systems, which can lead to increased efficiency and effectiveness (Idris and Durmuşoğlu, 2021).

Before *ISO 56000*, especially ISO 56002 and the new certified ISO 56001:2024, played an important role in setting the standards designed to provide a framework for Innovation Management Systems, there were other guidelines and methods developed in every country. For example, the British Standard BS 7000 Series, focused on design management, specifically addressing innovation management within the design process; AFNOR FD X50–271, a French national standard that provides a structured methodology for innovation management in businesses, especially for small and medium-sized enterprises (SMEs) and UNE 166000 Series (Spain), that focuses on R&D and innovation management, including certification standards for R&D project management (Mir et al., 2016).

In summary, the literature on Innovation Management Systems (IMS) has significantly advanced the field by providing comprehensive frameworks, identifying best practices, and addressing critical challenges organizations face in managing innovation. Key contributions include the development and refinement of standards like ISO 56002, which provide a structured approach for implementing IMS across diverse sectors. These standards have helped unify the understanding of innovation processes and set benchmarks for organizations striving to enhance their innovation capabilities. Overall, the literature has not only clarified the theoretical foundation of IMS but also provided practical tools and strategies for organizations to harness innovation systematically, driving both academic understanding and real-world application.

3. Research methodology

A systematic literature review (MacDonald, 2014; Snyder, 2019; Xia et al., 2024) was conducted by searching the Web of Science Database and Scopus Database, due to its wide range of academic disciplines indexing high-quality and reputable journals, conference proceedings, and research articles. This ensures that the content in the database is reliable and peer reviewed. A systematic literature review is instrumental in enhancing the study of Innovation Management Systems by offering a structured, comprehensive, and rigorous approach. It ensures that all the relevant literature is included, reducing bias in study selection and providing a fair and balanced representation of the field (Okoli, 2015). Additionally, a systematic review identifies gaps in the literature, guiding future research and preventing redundancy. Through its methodological rigor and evidence-based insights, it not only contributes to the credibility of a study but also informs decision-making and best practices in innovation management (Bracio and Szarucki, 2020).

A robust Systematic Literature Review (SLR) entails an initial phase of defining the research questions. Subsequently, the second step encompasses the formulation and implementation of an SLR protocol, followed by a deep analysis of the context, results, and future research, as explained in Fig. 2. For this study, the following questions were proposed to assess a more in-depth focus on the methodology, practical implications, cultural aspects, and sector-specific considerations associated with innovation management systems within the context of a literature review.

1. What are the main components and frameworks of innovation management systems discussed in the literature?

Fig. 2. Methodology. *Source: Adapted from (*Cannavacciuolo et al# 2023*).*

2. What are the latest trends and challenges in innovation management systems?
3. What gaps or inconsistencies exist in the literature, and what future research opportunities do they suggest?

Based on the initial approach, the first filter is applied: Innovation Management System was searched in all fields until December of 2023, resulting in 127,777 publications from Web of Science Core Collection. In order to get the articles corresponding with our search, the first restriction is applied. For the second filter, the search query was changed to the following equation, ("Innovation Management System") AND ("Innovation Management" OR "Innovation System"), resulting in 88 papers, which after a thorough reading of the articles and elimination of duplicates, was cut to 74 papers. In addition, the third filter added the search in WOS and Scopus database for the keywords "Innovation Management", "Literature Review", "Innovation Capacity" and "Innovation Capability", which resulted in 66 articles, of whom 44 articles were finally selected, in order to cover all the bases and be able to provide with an extensive review. Besides all the above, as shown in Fig. 3, all articles should be peer-review published papers and should be written in English, resulting in the final selection of 118 articles.

The main reason for the inclusion of the keywords "Innovation Capacity" and "Innovation Capability" is based on the fact that both terms have been used to identify the implementation of IMS, but it is really important for this study to recognize the difference between them to effectively understand how to carry out any innovative guidelines within IMS, especially since in the literature this type of discussion is becoming more frequent (Rajapathirana and Hui, 2018). Innovation capacity refers to an organization's ability to generate, adopt, and implement new ideas, processes, products, or services effectively. Innovation capacity is a multifaceted concept that goes beyond the mere generation of ideas to include the organization's ability to translate these ideas into tangible outcomes that create value (Cheng et al., 2023). Innovation capability refers to an organization's scope to systematically and consistently create value through the development and implementation of new ideas, products, processes, or business models. It goes beyond the idea generation phase to encompass an organization's ability to effectively manage, execute, and derive value from its innovation initiatives. In essence, innovation capacity represents the readiness and potential for innovation, while innovation capability extends to the organization's demonstrated proficiency in turning that potential into actual innovations.

A literature review with the research strategy as seen in Table 1, spanning two decades, not only offers a comprehensive understanding of the evolving strategies, methodologies, and best practices in innovation management but also serves as a valuable historical record of how organizations have adapted and thrived in an era of profound change. It provides the essential context for understanding the key developments and trends in innovation management and helps in identifying areas where further research and innovation are warranted. This period also covers the implementation of ISO 56002, which is incredibly important to include in this study, as it is the official standard to implement IMS.

A detailed descriptive and content analysis was conducted, resorting in various variables (such as distribution of articles per year and number of citations), and engaging in an in-depth examination of the chosen papers (Cannavacciuolo et al., 2023). Subsequently, a critical analysis and discussion of the results were managed, encompassing a thorough examination of the selected articles. To search into the forefront of literature on the main subject, the VOSviewer software program was used to study authors and bibliographic topics. The analysis considered key perspectives such as the time evolution, journals, authors, citations, research areas and common keywords.

4. Results

This study evolves with a meticulous examination of the key components and evolutionary trends in innovation management systems as documented in the literature. From the distribution of articles per year to citation indexes, the descriptive and content analyses offer a thorough overview, laying the groundwork for a refined exploration. By presenting the results it is expected to contribute not only to academic understanding of innovation management but also offers a practical guide for organizations seeking to unlock the full potential of their innovation management systems in the years ahead.

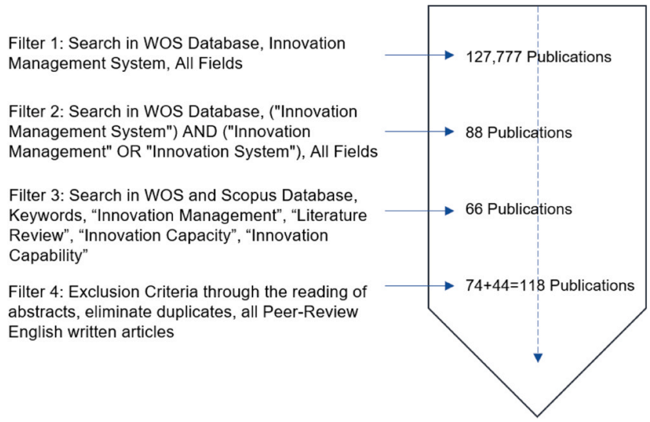

Fig. 3. Flow diagram of the literature selection process. *Source: Author's own contribution with data retrieved from the WOS database.*

Table 1
Research strategy.

Field	Option
Period	2003–2023
Main Database	WOS
Keywords Search Database	WOS-Scopus
Type of Documents	Articles and Conference Papers
Exclusion Criteria	Abstract, Keywords and Language

4.1. Overview and Bibliometric Analysis

The final selection of papers includes 118 contributions for which a bibliometric analysis was performed to provide a general overview of the investigated topic. 74 papers were selected for the search of “Innovation Management System”, seven for “Innovation Management”, 17 for the keywords of “Innovation Capacity”, and 22 for “Innovation Capability”. Only two articles were duplicated after the search, leaving 44. Fig. 4 shows the distribution of papers per year, in which 2017 and 2021 presented the most articles, with 12 each year, followed by 10 for the years 2015 and 2019. The temporal analysis of publications indicates a growing interest in the evolving trend of the subject. This underscores the increasing attention given to, and the perceived novelty of, the explored relationship between IMS and other research areas.

Besides the inclusion of proceedings and conferences, the articles selected appear to be highly multidisciplinary and published in a variety of journals. The most published journals are the Journal of Engineering and Technology Management, Technovation, Sustainability, the Journal of Business Research, and the International Journal of Innovation Management.

As seen in Table 2, the highest number of citations is registered in 2004, followed by 2011, 2014, and 2019. The trend suggests an increase in interest in covering the subject of IMS and the establishment of the framework and guidelines toward innovation. It is important to emphasize that Table 1 only covers the citations in WOS; therefore the inclusion of papers with no citations indicates either a private collection, a paid subscription to a journal, or a university collection. All the above concludes that just because a paper has zero citations in the WOS this does not mean that it should not be included in the study of IMS.

Table 3 shows the top ten most cited papers with the correspondent year. The most reported article is (Lüthje and Herstatt, 2004), which discussed The Lead User Method, an innovation management technique that involves identifying and involving "lead users" in the innovation process. Lead users are individuals or organizations that face needs similar to those of the broader market but are experiencing them in advance. This paper describes the early attempts to identify an Innovation Management System, as well as the capability for innovation and being ahead as an organization. The second most cited article is (Rohrbeck and Gemünden, 2011), which discussed the effects of implementing innovation and the capacity of a firm to successfully manage strategic management, innovation management, and the future agenda of the company. The third most mentioned paper is (Hollen et al., 2013), which leads the trend of IMS toward organizations and management, and how it improves resource productivity, which leads to higher environmental performance, which also increases firms’ competitiveness. It is important to highlight the review article (Khosravi et al., 2019) as one of the few review articles about Innovation Management, focusing more on managerial innovation empirical studies.

The following analysis presents the visualizations from "VOSviewer", a software tool developed by Nees Jan van Eck and Ludo Waltman at Leiden University’s Centre for Science and Technology Studies (CWTS). VOSviewer is designed for bibliometric analysis and visualization, and it is commonly used to explore and visualize relationships among scientific publications, authors, keywords, and other entities.

Fig. 5 presents the co-authorship map, by which nodes are the authors. 120 authors meet the threshold: a minimum number of one document for an author, and a minimum number of one citation for an author. The selected works are the result of the research effort of all authors, often collaborating in small, yet influential, research teams. With the exception of a few authors that produced two or more

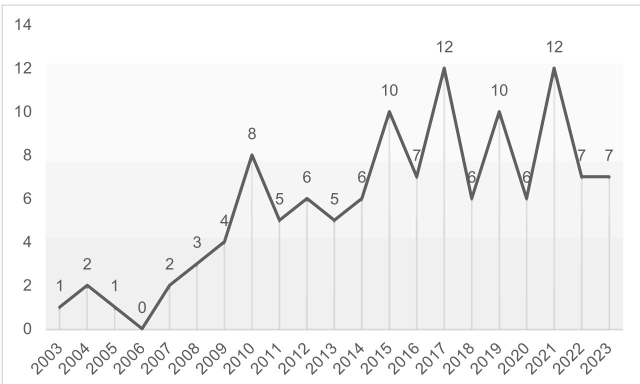

Fig. 4. Distributions of papers per year. *Source: Author’s own contribution with data retrieved from the WOS database.*

Table 2
Citations and number of articles per year.

Year	Number of Articles	Citations
2003	1	0
2004	2	218
2005	1	0
2006	0	0
2007	2	0
2008	3	2
2009	4	0
2010	8	47
2011	5	200
2012	6	106
2013	5	149
2014	6	150
2015	10	100
2016	7	110
2017	12	86
2018	6	42
2019	10	139
2020	6	16
2021	12	80
2022	7	41
2023	7	6

Table 3
Most influential authors and cited papers.

Rank	Authors	Title	Year	Journals	Citations
1	(Lüthje and Herstatt, 2004)	The Lead User method: an outline of empirical findings and issues for future research	2004	R & D MANAGEMENT	217
2	(Rohrbeck and Gemünden, 2011)	Corporate foresight: Its three roles in enhancing the innovation capacity of a firm	2011	TECHNOLOGICAL FORECASTING AND SOCIAL CHANGE	187
3	(Hollen et al., 2013)	The Role of Management Innovation in Enabling Technological Process Innovation: An Inter-Organizational Perspective	2013	EUROPEAN MANAGEMENT REVIEW	91
4	(Khosravi et al., 2019)	Management innovation: A systematic review and meta-analysis of past decades of research	2019	EUROPEAN MANAGEMENT JOURNAL	79
5	(Boly et al., 2014)	Evaluating innovative processes in French firms: Methodological proposition for firm innovation capacity evaluation	2014	RESEARCH POLICY	71
6	(Simon and Honore Petnji Yaya, 2012)	Improving innovation and customer satisfaction through systems integration	2012	INDUSTRIAL MANAGEMENT & DATA SYSTEMS	56
7	(Mir et al., 2016)	The impact of standardized innovation management systems on innovation capability and business performance: An empirical study	2016	JOURNAL OF ENGINEERING AND TECHNOLOGY MANAGEMENT	55
8	(Ferreira et al., 2015)	Drivers of innovation strategies: Testing the Tidd and Bessant model	2015	JOURNAL OF BUSINESS RESEARCH	54
9	(P. A. Khan et al., 2021)	Does adoption of ISO 56002–2019 and green innovation reporting enhance the firm sustainable development goal performance? An emerging paradigm	2021	BUSINESS STRATEGY AND THE ENVIRONMENT	52
10	(Dadfar et al., 2013)	Linkage between organizational innovation capability, product platform development and performance: The case of pharmaceutical small and medium enterprises in Iran	2013	TOTAL QUALITY MANAGEMENT & BUSINESS EXCELLENCE	51

papers *like Moises Mir, Martí Casadesús, Joao Pereira and Yuliia Klius*, the data shows that, in general, the identified research teams often produced only one co-authored paper in the area of study, thus suggesting that the organization of research on this topic is still evolving and not yet consolidated. The above clearly indicates that Innovation Management and IMS, even though an interesting and expanding research area, still needs to mature into a more development subject for more authors to join the exploration of innovation on organizations and the competitive business landscape, that leads to a sustained success within the field.

For the visualization, Fig. 6 presents the Co-occurrence Keyword Map, with the most used keywords from the selection of articles. The size of each item is related to the frequency with which it occurs in the selected sources. As expected, “Innovation”, “Innovation Management” and “Innovation Management System” are the most used keywords, followed by “Knowledge Management”, “Total Quality Management” and “Management”. Given that the first three are highly integrated with one another and with “Management”, its more worthy to highlight that “Total Quality Management” and/or “Quality Management” is a very interesting keyword to analyze, due to the fact that TQM and IMS focus on different aspects, even if they can be complementary concepts, too much distinct on innovation without proper quality controls may lead to the introduction of subpar products. Conversely, excessive rigidity in quality

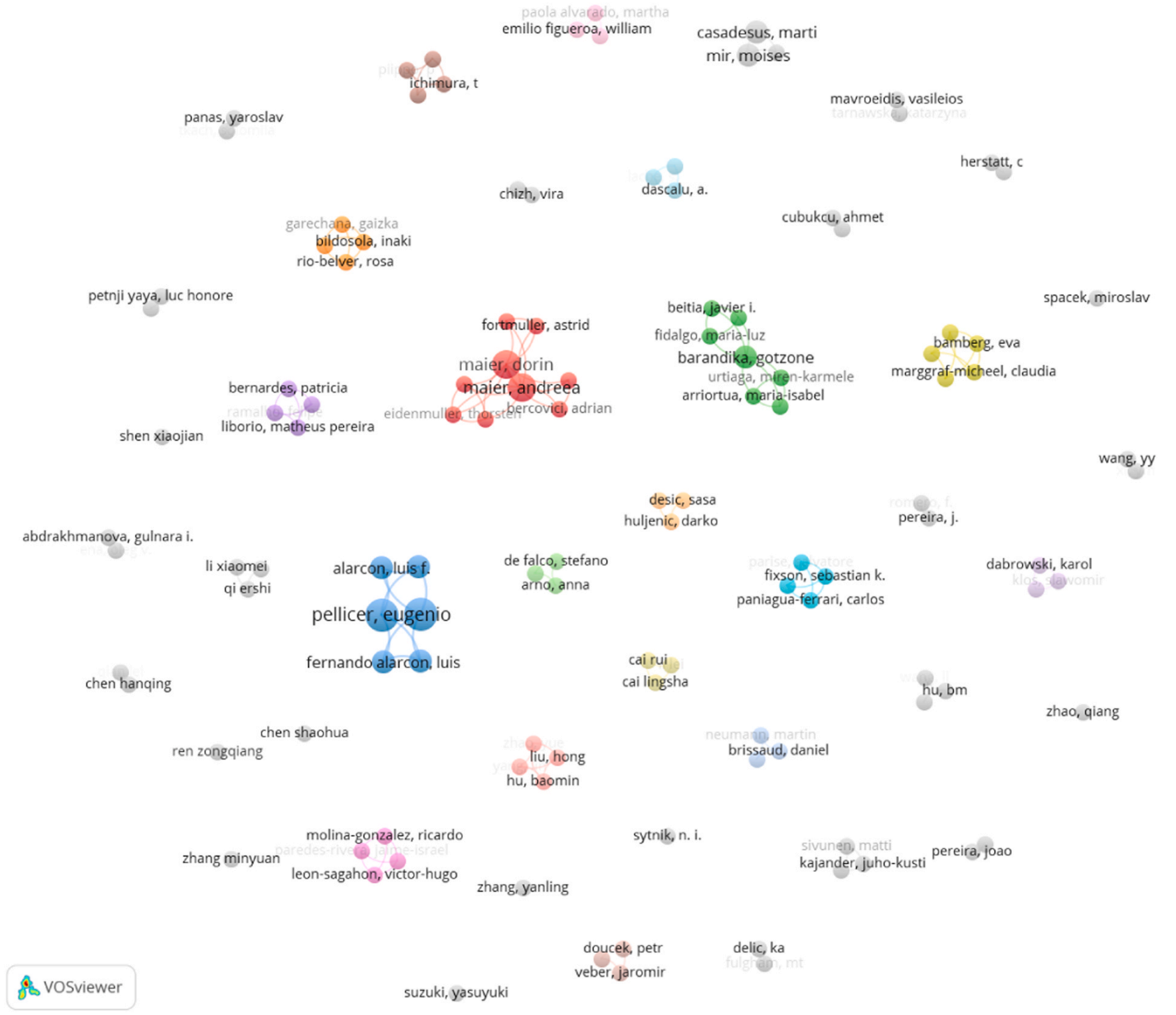

Fig. 5. Co-authorship map *Source: Author's own contribution using VOSviewer.*

management without room for innovation may hinder an organization's ability to adapt and stay competitive (García-Fernández et al., 2022). In the case of "Knowledge Management", Knowledge Management and Innovation Management are intertwined concepts, each reinforcing and enhancing the other. An effective KM system creates an environment where knowledge flows seamlessly, contributing to the organization's innovation capabilities and overall success (Zámborský et al., 2023).

Despite the inclusion of "Innovation Capacity" and "Innovation Capability" in the search for the articles to this study, these keywords do not seem to be the most used ones, mainly because authors prefer to use the term "Innovation" or "Innovation Management" to identify their research, even is the subject for the investigation relies on the above. It can also be important to mention that is not common for some papers that provide the same area of study to use the same keywords, which can cause a problem when try to research a specific area, as WOS or other databases cannot find the articles needed for the study.

4.2. Content Analysis of the Sources

In Section 4.1 *Overview and Bibliometric*, we provide a broad examination of the literature, using quantitative methods to analyze publication trends, citation networks, and key authors, journals, or keywords. This helps establish the scope of the field, highlight influential works, and identify research clusters. The following section ensues logically by diving deeper into the qualitative aspects of the literature. While bibliometric analysis maps the structure and impact of research, content analysis interprets the themes, methodologies, and findings within the reviewed studies. This transition allows for a more comprehensive understanding of the literature, moving from a high-level mapping of research activity to an in-depth exploration of key insights, trends, and gaps in innovation management systems.

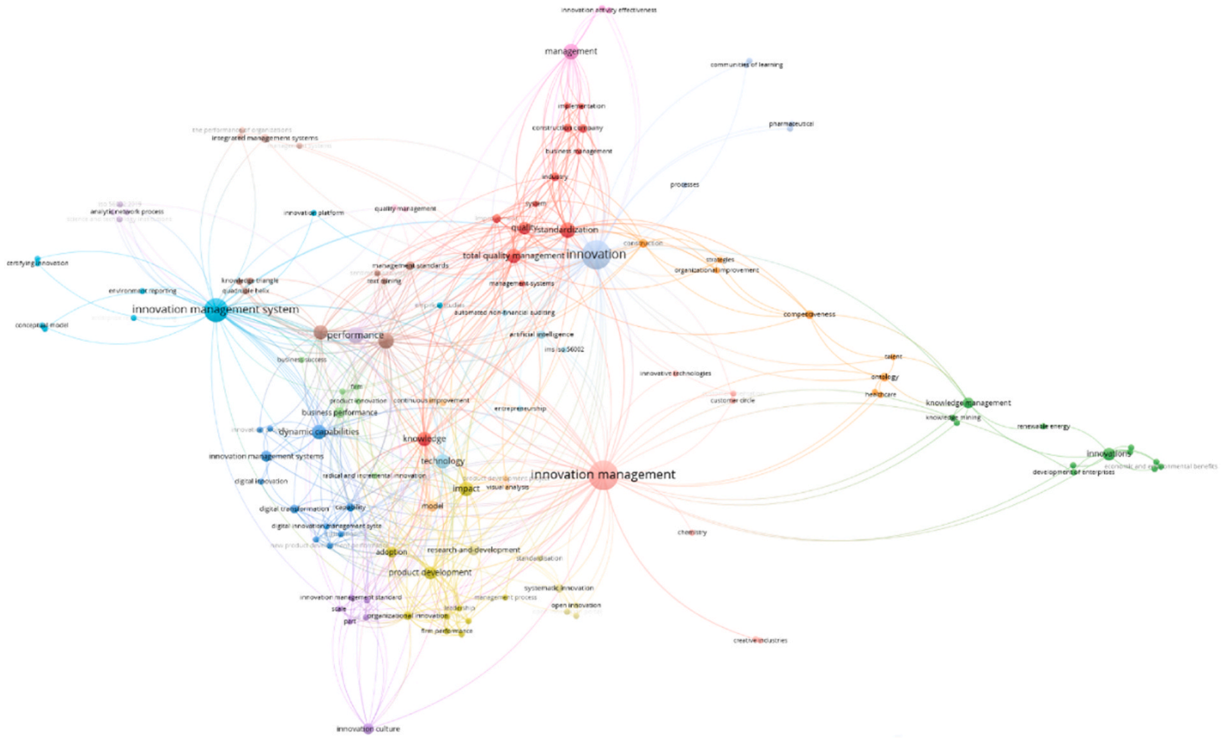

Fig. 6. Co-occurrence keywords map. *Source: Author's own contribution using VOSviewer.*

For this section, a rigorous content analysis is conducted, after the methodology found in (Cannavacciuolo et al., 2023). According to (Bryman, 2014), content analysis is a systematic research technique used to analyze textual, visual, or audio data by categorizing and quantifying patterns, themes, and meanings within the content. It involves systematically examining and evaluating the content of the selected sources, which aims to identify key themes, concepts, patterns, and relationships within the literature related to the research topic or question. As describe in Fig. 7, this process includes four steps: *Data Collection, Bibliometric Analysis, Cluster Interpretation and Labeling, and Final Clusters*. The first step presents relevant academic articles on Innovation Management Systems (IMS) that were retrieved from Scopus and Web of Science within the timeframe 2003–2023, resulting in 118 articles. The selection was based on citation count, relevance to IMS, and peer-reviewed status to ensure high-quality sources. For step 2, using VOSviewer, a bibliometric analysis was conducted through bibliographic coupling, which grouped articles based on shared references, authors and keywords. For step 3, each cluster was examined through qualitative content analysis, identifying recurring themes, topics, and keywords. Articles within each cluster were reviewed to ensure coherence, and appropriate labels were assigned based on their dominant subject matter, ensuring meaningful categorization.

Finally, step 4 shows that the content of the contributions can be grouped in four main macro topics or clusters: *Theories and Conceptual Frameworks, Methodological Approaches, Research Areas & Sectors and Current and Future Impacts*, as shown above:

1. Theories and Conceptual Frameworks: Covering foundational theories, definitions, and models related to IMS.
2. Methodological Approaches: Addressing research methods, tools, and evaluation techniques applied in IMS studies.

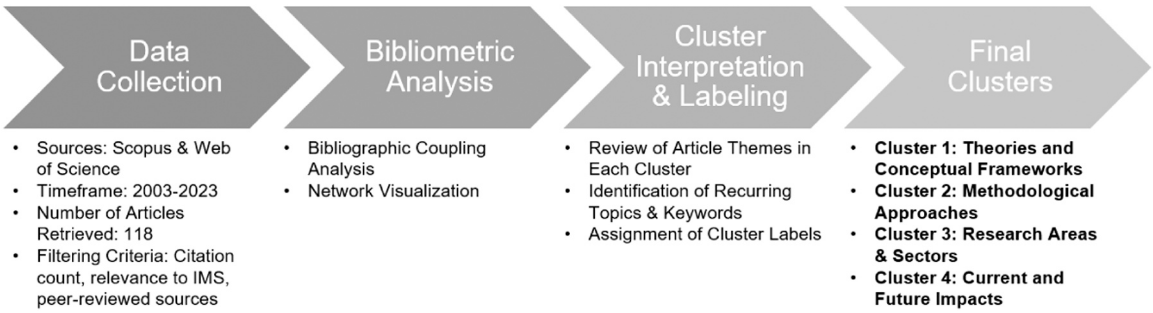

Fig. 7. Formation of Clusters.*Source: Author's own contribution.*

3. Research Areas & Sectors: Identifying industry-specific and sectoral applications of IMS.
4. Current and Future Impacts: Discussing emerging trends, challenges, and the evolving role of IMS in organizations.

In the initial cluster, the primary theories and conceptual frameworks employed in the selected articles are examined and analyzed. The second cluster encompasses the methodological approaches employed in the articles, including literature reviews and case studies, among others. The third cluster categorizes arguments related to the research areas or sectors investigated in the articles, such as manufacturing, education, management, etc. Finally, the fourth cluster delves into the present and future impacts of Innovation Management Systems (IMS) and outlines the prospective direction of the main topic. Subsequently, the following section presents and scrutinizes the findings within these four clusters.

Each cluster was named based on the dominant themes and recurring topics identified within the grouped articles. We now provide clearer justification for these labels by linking them directly to key studies within each category.

4.2.1. "Theories and Conceptual Frameworks" Cluster

This cluster focuses on theories and conceptual frameworks, which systematically examine and categorize the prevalent theoretical foundations and conceptual models within the selected sources. The process involves creating specific categories aligned with the research objectives, and each source is meticulously analyzed to identify and code instances where particular theories or conceptual frameworks are employed. This cluster includes 55 articles that discussed a theory or a conceptual framework, divided into nine categories.

The first category is *Innovation Management System (IMS)*, which includes articles, conferences, and proceedings that discussed the topic within various theories and conceptual frameworks that guide organizations in systematically fostering and managing innovation, as shown in Table 4. This selection is focused on both the discussion or application of IMS within various areas. In the context of business and organizational management, (Chanal, 2004; Desic et al., 2007; M. Zhang, 2010; Wang et al., 2011) focus on implementing innovative ideas within an enterprise, with the latter two also incorporating Product Innovation Management in the same context. (Lendel et al., 2015; Badrinas and Vilà 2015; Maier et al., 2015; Peterková et al., 2020; Damodharan et al., 2022) discussed how the

Table 4
Selection of articles for the first category of Cluster 1.

Title	Year	Main Findings
Building up an enterprise technological innovation management system based on project management: A new approach to system analysis	2003	A new approach to solving the problem in enterprise modeling or system analysis on how to make the model flexible.
The Lead User method: an outline of empirical findings and issues for future research	2004	Exploring the Lead User Method in innovation and marketing.
Innovation management as a basis for successful development	2007	How to develop and maintain a company culture favoring innovation.
Innovation Management and Organizational Learning: a Discursive Approach	2009	How to design innovative organizations through theoretical and managerial insights.
A Chief Technology Officer's Views and Behaviors in the Dual Innovation Management System	2009	A discussion of the concept and approach of the Dual Innovation Management System.
Study on the Design of an Enterprise Product Innovation Management System	2010	How to combine product innovation management with the existing product operating management.
An Approach for Lean Product Innovation Management	2011	How to implement Lean Management to promote the innovation capability of an enterprise.
Introduction to Process Innovation Management in the Process Map of a Materials Science Research Group: Influence on the Formation of Doctorates	2014	How to extend innovation to university research groups.
The new energy industries' core competencies strategy based on knowledge innovation management	2014	Explores the relationship between knowledge innovation management and enterprise core competencies.
Management of Innovation Processes in Company	2015	How to apply IMS and create a model in a company.
An Innovation Management System to Create Growth in Mature Industrial Technology Firms	2015	Exploration of the creation of significant innovations with value within a company.
Innovation as a part of an existing integrated management system	2015	Innovation Management in the context of an existing integrated management system implemented in an organization.
Development of knowledge management infrastructure in an organization	2016	Analyze development of KM infrastructure at various stages of knowledge management evolution at the organization.
Development and Operationalization of a Model of an Innovation Management System as Part of an Integrated Quality-Environment-Safety System	2017	Develop a model of IMS as part of an integrated management system.
Toward a New Innovation Management Standard. Incorporation of the Knowledge Triangle Concept and Quadruple Innovation Helix Model into an Innovation Management Standard	2017	Standardization of IMS with the Quadruple Innovation Helix model and the Knowledge Triangle concept.
Business Models for Corporate Innovation Management: Introduction of a Business Model Innovation Tool for Established Firms	2018	Incorporation of a BMI in a firm in the context of innovation.
Integrated AI and Innovation Management: The Beginning of a Beautiful Friendship	2020	How AI will support various aspects of innovation management.
Evaluation of innovation activities and the innovation management model of selected innovative companies	2020	Evaluation of IMS within selected companies.
The Innovation Management Footprint in a Technology Organization	2022	Analysis of the efficiency and effectiveness of IMS.

relationship between IMS and enterprises is critical for fostering a culture of innovation and ensuring that innovative initiatives align with the organization's strategic goal.

(Rui et al., 2014; Sytnik, 2015) discussed the concept of Knowledge Management, a holistic strategy that extends beyond technology implementation; it encompasses cultural, organizational, and process-oriented aspects to create a learning and adaptive environment that promotes innovation. The following articles consider different theories; (Hu, Liu, and Wang, 2003) relates the enterprise management system to a new system analysis, (Suzuki, 2009) discussed Dual Innovation Management System, (Maier et al., 2017) integrates a model of IMS with a Quality-Environment-Safety System, (Mavroeidis and Tarnawska, 2017) with Quadruple Innovation Helix model and the Knowledge Triangle concept, (Trapp et al., 2018) incorporate Business Models with Innovation (BMI), (Yams et al., 2021) combines IMS AND AI, and finally (Barandika et al., 2014) on how to extend innovation to university research groups.

The second category is *Innovation,* as seen in Table 5, which includes 8 articles and proceedings that solely discusses the concept of innovation, a complex and multifaceted idea informed by various theories and perspectives. Different theories contribute to the understanding of the mechanisms, drivers, and outcomes of innovation. Most of the articles selected integrates innovation with different areas, but stay focused on the range of the scope of this review, meaning that they must include IMS within the research. (Delic and Fulgham, 2004; Palm et al., 2016; Pihlajamaa, 2017) study innovation in terms of different environments and how to processes them, (Simon and Honore Petnji Yaya, 2012; Gloet and Samson, 2014) focuses on performance, while (Çubukcu and Gümüş 2015; Y. Zhang, 2015) and (Stevanovic et al., 2016) considers other topics such as tools, education and development.

The third category is *SMEs*, which includes articles and conferences as shown in Table 6, describes mostly the relationship between an Innovation Management System (IMS) and Small and Medium Enterprises (SMEs), which is crucial for fostering a culture of innovation and ensuring sustainable growth. SMEs, with their limited resources and unique challenges, can benefit significantly from implementing an effective IMS. From Table 6, (Silva et al., 2018; Pereira and Romero, 2018) details on how to implement an IMS in SMEs, by optimizing their resources by providing a structured framework for identifying, prioritizing, and implementing innovative ideas. This ensures that resources are directed toward initiatives with the highest potential for impact. (Ren, 2009) and (Z. Xu, 2018) integrates a new system and model to improve innovation regarding capability and customer service in SMEs.

Table 7 includes the fourth category which integrates *Framework* and *Performance,* that contains a range of topics related to conceptual frameworks, theoretical models, and empirical studies that examine the performance outcomes of various frameworks. Both concepts relate to most articles, regarding the analysis of an enterprise performance, like the following papers (Dervitsiotis, 2010; Maier et al., 2017; Albors-Garrigos et al., 2018), focus on an more digital impact (Santarsiero et al., 2020; Z. Tian and Wang, 2022) or an approach to innovation capability (Daronco et al., 2023).

The fifth category in Table 8 includes the study of *Innovation Capacity and Innovation Capability*; the first one represents the broader potential and readiness for innovation within an organization, while the second one is the specific ability to execute innovation-related activities effectively. Together, they form a dynamic and interrelated framework that allows organizations to not only possess the potential for innovation but also to realize tangible outcomes and stay competitive in a rapidly changing business environment. Regarding the articles that discussed innovation capacity, (Rohrbeck and Gemünden, 2011; Okatan, 2017; Okatan and Alankus, 2021; Rocha et al., 2023), present their findings in regards of enhance innovation in a corporate level. (C.-M. Alexe, 2009; George,et al., 2012; Xu, Wu, and Ren, 2012; Djoumessi et al., 2019; Chen, Yin, and Li, 2020) do the same for innovation capability in a firm, with the latest moving forwards sustainability.

Finally, the sixth category is *Technologies,* as shown in Table 9, which proposes the integration of IMS with technologies an ongoing process, and organizations need to stay abreast of emerging technologies to leverage their full potential in fostering innovation. The goal is to create a technology-enabled ecosystem that supports the entire innovation lifecycle, from idea generation to implementation and evaluation. (Popescu et al., 2011; Hollen et al., 2013) study technological changes through IMS, while (Endres et al., 2022) includes a software to support IMS, and (Li and Hai, 2023) the benefits to include a more sustainable perspective.

4.2.2. *"Methodological Approaches" Cluster*

The second cluster is "Methodological Approaches", which contains 21 articles, divided into case studies and literature reviews.

Table 5
Selection of articles for the second category of Cluster 1.

Title	Year	Main Findings
Corporate innovation engines: Tools and processes	2004	How to incorporate IP and processes behind innovation activities.
Improving innovation and customer satisfaction through systems integration	2012	How innovation and customer satisfaction relates to a management system.
Managing Knowledge and Innovation for Performance	2014	How knowledge and innovation management practices contribute to innovation performance.
Systematic Design of an Open Innovation Tool	2015	Innovation and "open innovation portals".
Research on Innovation for Higher Education Management	2015	IMS and higher education in China.
The challenge of integrating innovation and quality management practice	2016	The potential integration of innovation and quality management practice within the public sector.
Ideas Assessment and Selection in Product Innovation - The Empirical Research Results	2016	Assessment for product development and innovation.
Going the extra mile: Managing individual motivation in radical innovation development	2017	How to manage individual motivation in radical innovation development.

Table 6
Selection of articles for the third category of Cluster 1.

Title	Year	Main Findings
The Complexity of Total Innovation Management System in SMEs	2009	How to integrate a systemic capability for innovation in SMEs
The Two Wheel Model: A Reference for the Innovation Management System in Small and Medium Enterprises	2018	A new model for IMS in SMEs
Small and Medium Enterprises Innovation Management System Based on Clustering Algorithm	2018	How to manage customer service in SMEs
An Innovation Support System for SMEs	2018	How to support the management of innovation in SMEs
A Research, Development and Innovation Management System for Small and Medium Enterprises	2018	IMS for SMEs

Table 7
Selection of articles for the fourth category of Cluster 1.

Title	Year	Main Findings
A framework for the assessment of an organization's innovation excellence	2010	How IMS facilitates a firm's adaptation to new conditions.
Innovation Management System - a Necessity for Business Performance	2017	Analyzes the main models of innovation management systems.
Innovation Management Techniques and Tools: Its Impact on Firm Innovation Performance	2018	IMTS on innovation performance of firms.
Innovation Labs to Foster Digital Innovation: A Management Framework	2020	IMS and the role of Innovation Labs.
Construction of an enterprise innovation performance model using knowledge base and edge computing	2022	Implementation of the knowledge value chain (KVC) model.
A new framework of firm-level innovation capability: A propensity-ability perspective	2023	A systematic approach to formulate a firm-level IC framework

Table 8
Selection of articles for the fifth category of Cluster 1.

Title	Year	Main Findings
The Analysis of Innovation Capability at the Level of a Firm	2009	The innovation capability of a firm depends on the ability of management.
Corporate foresight: Its three roles in enhancing the innovation capacity of a firm	2011	How foresight should be integrated with the innovation effort of a company.
The Model of Analysis and Improvement of the Firm's Innovation Capability	2012	How the innovation capability of a firm depends on the ability of the managers.
Synergy of Innovation Elements and Evolution of Innovative Capability-With System Dynamics Modeling	2012	How innovative capability is vital to firm's competitive success.
Effect of Organizational Culture on Internal Innovation Capacity	2017	Correlation between the dimension of the internal innovation system and organizational culture.
Deconstructing Lawson and Samson's Concept of Innovation Capability: A Critical Assessment and a Refinement	2019	Discussion on innovation capability in organizations with a dynamic capabilities approach.
Firm innovation system: Paths for enhancing corporate indigenous innovation capability	2020	This study proposes four systematic paths for improving the firm innovation system.
Effect of External Innovation Capacity on Company Level Innovativeness	2021	Discussion of the factors affecting the performance of organizations.
Measuring and Evaluating Organizational Innovation Capacity and Performance from a Systemic and Sustainability-Oriented Perspective	2023	Proposition of a conceptual model for measuring and evaluating the sustainability-oriented innovation capacity.

Table 9
Selection of articles for the sixth category of Cluster 1.

Title	Year	Main Findings
Technological Changes and Quality Management Relationship	2011	Possibilities of integrating technology change management and quality management.
The Role of Management Innovation in Enabling Technological Process Innovation: An Inter-Organizational Perspective	2013	How to improve resource productivity and environmental performance.
Digital innovation management for entrepreneurial ecosystems: services and functionalities as drivers of innovation management software adoption	2022	The inclusion of software to support innovation management methods and activities.
Evaluation of Economic Security and Environmental Protection Benefits from the Perspective of Sustainable Development and Technological Ecological Environment	2023	Implementation of eco-technology IMS.

Both are complementary approaches that offer unique contributions to the understanding of IMS. While case studies provide insights from real-world implementations, the literature reviews synthesize existing knowledge, identify trends, and inform future research and practice.

Table 10 shows the first category, *Literature Review*, with only four existing articles since 2019, which coincides with the start of ISO 56002, Innovation management system – Guidance. (Khosravi et al., 2019) manage a systematic review and meta-analysis of the literature into existing empirical studies on management innovation. The content analysis reviews over 66 studies from 1981 to 2017, analyzing the trends and background of research into management innovation. (Solaimani et al., 2019) delves into Lean Innovation Management with a systematic literature review. This paper elaborates on how the proposed framework can be applied, with a total of 88 publications for the review, leading to 34 Lean principles and practices relevant to innovation management. (Idris and Durmuşoğlu, 2021) contributes theoretically to the development of literature on IMS and provides a clear understanding of the state of the field during the period 2006–2020, with 73 articles selected for the study. (P. Tian et al., 2021) focus on product development project through the years 2012–2021, but with an approach to innovation management, system modelling and knowledge management of development organization, hence the inclusion in this study.

The second category is *Case Studies,* with 17 articles selected, as seen in Table 11. This group is important for the application of IMS, as enhances the richness and applicability of the review by grounding theoretical concepts in real-world experiences. This approach not only strengthens the academic understanding of IMS but also provides valuable insights for practitioners seeking to implement or improve innovation management within their organizations. Case studies offer rich and detailed information about the intricacies of IMS implementation, including the challenges faced, decisions made, and outcomes achieved. This level of detail allows for a nuanced understanding that goes beyond broad theoretical principles. (Qin and Feng Dong, 2007) and (Mao et al., 2008) discussed IMS in the automotive industry in China, with the first one applying the concept of innovation capacity and the later in regards of the product and a more collaborative innovation. (Chen, 2010) focuses on the study of the construction industry in China and how to apply IMS, (Xu, J and Xian, 2013) while develops a model towards SMEs in West China. (Pellicer et al., 2014) work with the construction industry and the implementation of IMS, (Molina-González et al., 2010) gives an ontology approach includes IMS within a solar technology company, (Saidi et al., 2021) with IMS and the health care sector and (Chapa et al., 2023) with the tourism sector. Finally, (Shen, 2010) and (Mir et al., 2016) present an empirical study of Product Innovation Management and the standardization of IMS, respectively. (Shi, Xu and Peng, 2013) presents a study of IMS and small cities soft power, (Zeledon-Ruiz et al., 2017) implement a strategy in higher education through IMS, (Ferreira et al., 2015) studies innovation capacity within an IMS structure and (Gutiérrez et al., 2021) at a water supply company in Bogotá, Colombia.

4.2.3. *"Research areas and sectors" Cluster*

Table 12 presents the third cluster, "Research areas and sector", which contains 21 articles divided into different industries. This section helps to improve understanding of IMS by providing a contextualized view of the challenges, opportunities, and dynamics specific to that industry. Also, organizations can anticipate potential disruptions within their industry and IMS can be designed to proactively address and capitalize on these disruptions, ensuring organizational resilience.

Companies play a crucial role in the economy by creating jobs, driving innovation, and contributing to economic growth. Within the concept of an organization, company or enterprise, over seven articles studied the subject. (Lei and Hanqing, 2008) delves into the concept of creativity within an innovation framework, (Smith et al., 2017) with regard to entrepreneurship, (Garechana et al., 2017) study the effect of standardization of an innovation system and data mining, and (Klius and Chizh, 2017) work with the creation of a "corporate innovation management chart" at industrial enterprises. (Panas and Tkach, 2017) research the analysis of the effect of IMS on enterprises, (Miroshnikov, Morozova, and Svetlichnaya, 2019) integrate quality management, and (Klius et al., 2021) introduce an interactive approach to IMS.

The construction industry is a sector of the economy that encompasses the planning, design, development, construction, and maintenance of buildings and infrastructure, and the interest in including innovation and IMS has increase over the years. (Pellicer et al., 2012) studies standardized innovation management system in a medium size construction firm, while (Yepes et al., 2016) the effects of implement IMS on a Spanish construction firm. The pharmaceutical industry plays a pivotal role in advancing medical science and improving health outcomes, contributing to the well-being of individuals and populations globally. Regarding this industry, (Spacek, 2012) works with the application of Stage-Gate Control Process (SGCP) within IMS, (Dadfar et al., 2013) with SMEs in Iran in the same industry, (Robb et al., 2022) focuses more in the economic models in healthcare and (Moreno-Conde et al., 2019) within the ontology sector, which often refers the formal representation of knowledge within a specific domain, presented as a

Table 10
Selection of first category of Cluster 2.

Title	Year	Main Findings
Management innovation: A systematic review and meta-analysis of past decades of research	2019	A systematic review and meta-analysis of empirical studies on management innovation.
On the application of Lean principles and practices to innovation management A systematic review	2019	Integrative view on Lean Innovation Management, through a systematic review.
Innovation Management Systems and Standards: A Systematic Literature Review and Guidance for Future Research	2021	A systematic review on IMS/Standards.
A Bibliometric Review and Visualization Analysis in Product Development Project Management: 2012–2021	2021	Literature Review on product development project and Innovation Management.

Table 11
Selection of second category of Cluster 2.

Title	Year	Main Findings
The innovation strategy of automotive independent brand under opening condition - Case study of brilliance innovation management	2007	How to promote innovation capacity in the automotive industry in China.
Study on the Collaborative Innovation Management Model of Automobile Complex Products Based on Product Total Lifecycle	2008	The innovation management system of Chinese automotive manufacturing enterprises.
Study on Construction of China's Industrial Technology Innovation Management System	2010	Study of industrial technology innovation operation mechanism.
Ontology-Based Integration of Training Systems in the Electrical Power Production Operators Domain	2010	An ontology-based approach to training management systems.
Empirical Study on Product Innovation Management Theory	2010	Introduction of new product innovation concept.
Study on the Technological Innovation Management Mode and Optimized Path of Resource-based SMEs in West China Based on Capability	2013	Study of technological innovation management of resource-based SMEs in West China.
Based on Innovation Management to Improve Underdeveloped Small Cities' Soft Power	2013	Study of IMS to enhance the City's soft power.
Model for Systematic Innovation in Construction Companies	2014	Case study of innovation in a construction company.
Drivers of innovation strategies: Testing the Tidd and Bessant model	2015	Study on innovation capacity within an innovation management structure.
The impact of standardized innovation management systems on innovation capability and business performance: An empirical study	2016	The impact of a Standardized Innovation Management System (SIMS) on a company.
The Dilemma of Innovation in the Construction Company: A Decade of Lessons Learned	2017	Study of a standardized innovation management system in a mid-size Spanish construction company.
Developing a project delivery system for a construction project – a case study	2017	Innovation Management System for Construction Sector Companies.
An Innovation Model for EPC/Turnkey Sector: The Case of Abengoa Solar New Technologies	2019	Case Study of an innovation model put into practice at ASNT.
From Self-evaluation to Quality Management and Innovation. The case of the School of Business Administration (UCR)	2019	Evolution of quality management and innovation as strategic on UCR.
Making 'hidden innovation' visible? A case study of an innovation management system in health care	2021	Study of the innovation monitoring and management system in the Norwegian health care sector.
Communities of learning as support for one knowledge and innovation management system: A case study	2021	Innovation Management System at Bogota's Water Supply and Sewage Company
Technology and innovation in intelligent tourist destinations. Case: Cuenca, Ecuador	2023	Study on the innovation management system in entrepreneurship.

hierarchical structure of concepts and their relationships.

Finally, both (Cerezo-Narváez et al., 2019) and (Chebukhanova and Zimakov, 2022) work within the aerospace industry, (Furjan et al., 2020) and (Gordeeva et al., 2020) on digital innovations and effectiveness in the sector of agriculture, (Peihong and Jiaqiong, 2009) analyses a more technical model on manufacturing, (Neumann et al., 2013) on product development in the automotive industry, (Lazarenko et al., 2019) works with the maintenance of innovation in mining companies, and (Robb et al., 2022) with renewable energies and IMS.

4.2.4. "Current and Future Impacts" Cluster

Assessing both current and future impacts on Innovation Management Systems (IMS) serves as a foundational step in understanding the practical implications and strategic relevance of IMS in contemporary business environments. By bridging the gap between theory and practice, understanding current impacts guides organizations in their pursuit of continuous improvement and enhances their competitiveness by leveraging successful strategies and innovations observed in the contemporary implementation of IMS.

This cluster is divided into two categories, the first being "Implementations of IMS", with 15 articles, as shown in Table 13. Examining implementations provides insights into how IMS concepts and frameworks are applied in real-world organizational settings. It bridges the gap between theoretical models and practical strategies, offering a nuanced understanding of IMS in action. (Q. Xu et al., 2008; Fang et al., 2010; C.-M. Alexe and Scarlat, 2010; Marggraf-Micheel et al., 2010; Boly et al., 2014; Golja and Kontosic, 2015; Kopkas, 2017) study innovation implemented in SMEs in different regions, with a focus on performance and improvements related to Innovation Capability and Innovation Capacity. (Nilsson and Ritzén, 2014) work with empirical investigations, (Fernandes et al., 2015) in improving the capabilities of a KIBS firm, (Carpio et al., 2015) study IMS and a Triple Helix Model in Ecuador, and (C.-G. Alexe and Alexe, 2016) implement an evaluation of Innovation Capability based on scores. Finally, (Klos et al., 2016) relates an ERP with IMS for the development of new products while (Boly et al., 2022) works with Innovation Capacity to improve cities' administrations at a local level, and (Sarsen et al., 2023; Malon, 2023) proved the implementation of IMS.

Table 14 presents the second category is "ISO and Standards". This selection presents the previous work related to IMS and ISO 56002, as long as the adoption of the standard since 2019. The ISO 56002 is a standard that provides guidelines for the establishment, implementation, maintenance, and continual improvement of an Innovation Management System (IMS). The importance of ISO 56002 to IMS lies in its role as a comprehensive framework that offers guidance to organizations seeking to enhance their innovation management practices. (Mir-Mauri and Casadesús-Fa,)1; Barandika et al., 2015 (Martínez-Costa et al., 2019) , (Giménez-Medina et al., 2023) work with European standards such as UNE 166.000 and UNE 166002: 2006, which represented previous version before the official guideline for ISO 56000 was develop for IMS. (Arnò1 et al., 2012) studies a proposition for a IMS Certification, (Khan et al.,

Table 12
Selection of articles for Cluster 3.

Title	Year	Main Findings
Technology innovation management system analysis based on two kinds of organization model in manufacturing industry	2005	Analyzes the technical innovation management system model.
Study on Innovation Management from the Perspective of Creative Industries	2008	Comparison and difference of the concepts of innovation and creativity.
Organizational Improvement Through Standardization of the Innovation Process in Construction Firms	2012	Innovation in a medium-size construction firm with a standardized innovation management system.
Stage-Gate Control Process and its Creative Adaptation to the Management of Innovations in Generic Pharmaceutical Business	2012	Application of Stage-Gate Control Process (SGCP) within IMS.
Linkage between organizational innovation capability, product platform development, and performance: The case of small and medium pharmaceutical enterprises in Iran	2013	Innovation in small and medium pharmaceutical enterprises (SMEs) in Iran.
IT-supported innovation management in the automotive supplier industry to drive idea generation and leverage innovation	2013	Innovation management within the automotive supplier industry.
Creative Innovation in Spanish Construction Firms	2016	The implementation of IMS within a Spanish construction firm.
The Evolution of an Innovation Capability Making Internal Idea Competitions Work in a Large Enterprise	2017	IMS approach to innovation and corporate entrepreneurship.
Effects of innovation management system standardization on firms: evidence from text mining annual reports	2017	The effects that standardization have in attitudes and values as regards innovation.
Strategic Analysis of the Effective Development of Industrial Enterprises on the Basis of the Use of Corporate Innovation Management Chart	2017	The use of a "corporate innovation management chart" at industrial enterprises.
The Features of Innovation Management at Ukrainian and European Enterprises	2017	Systems of innovation management at Ukrainian and European enterprise
Standardizing Innovation Management: An Opportunity for SMEs in the Aerospace Industry	2019	Adoption of R+D+i management system.
ITEMAS ontology for healthcare technology innovation	2019	Study on ITEMAS innovation indicators and innovation management system standards.
Toward an Integrated Approach to Improving Innovation Management System of Mining Companies	2019	Study on maintaining a company's innovation practices on the mining sector.
Integrating the Innovation Management Methodology into the Existing Quality Management System of a Machine-Building Enterprise	2019	How to integrate innovation management methodologies into the systems of quality management.
Value of Innovation Platforms in Agriculture	2020	Management of digital innovations in agriculture.
Management of Innovation in the Agricultural Sector	2020	Effectiveness of innovation in the agriculture industry.
Knowledge and Innovation Management for Transforming the Field of Renewable Energy	2021	The application of methods and tools of knowledge and innovation management at renewable energy enterprises.
Development of a Calculation Basis for the Goals and Objectives of Innovation Management at Industrial Enterprises in the Context of Post-Conflict Transformation	2021	The economic impact of introducing an integrative approach to innovation management in industrial enterprises.
Resource support of innovative small and medium-sized enterprises for space industry development in Russia	2022	Innovative small and medium-sized enterprises (SME) for the development of the space industry in Russia.
Enhancing organizational innovation capability - A practice-oriented insight for pharmaceutical companies	2022	Enhance pharmaceutical innovation within a context of changing economic models in healthcare.

2021) works towards sustainability, (R. Khan, Adi, and Hussain, 2021) on how to audit innovation with AI, while (S. B. da Silva, 2021) and (Santos et al., 2022) implements the adoption of ISO 56002 2019.

5. Discussion and Research Agenda

The purpose of this study is to construct a systematic literature review on the Innovation Management System, within the context of the potential and implementation of an IMS. The main findings reveal a growing interest in the subject, as over 118 articles study IMS, while significant advances have been achieved on the field in different sectors and enterprises. All the contributions were divided into four clusters, 1) Theories and Conceptual Frameworks, 2) Methodological Approaches, 3) Research Areas & Sectors and 4) Current and Future Impacts, as seen in Fig. 8, with the number of articles of each cluster presented as well. This categorization is very important for this study, as it shows how the literature has been studied and selected into different areas of research, and it helps in finding the main trends and gaps of the subject of study in the present article. A very important highlight, as seen in Fig. 5, the co-occurrence keywords map, is the establishment of a common ground in terms of keywords for "Innovation Management System", as well as the definition, as a framework or set of processes designed to manage and enhance innovation within an organization. Since the family of ISO 56000 was implemented in 2019, there is a common foundation for innovation within an organization, since the standard incorporates best practices and provides detailed guidance on establishing and maintaining an effective IMS.

Based on the findings, current research has been mainly oriented to the foundational frameworks and principles of IMS, with over half of the selected literature focusing on the conceptual aspects of IMS, rather than explore deeper into sector-specific applications, as seen in Tables 4–9 and Fig. 8. Theorical knowledge stills prevail over real-world application, with a variability in IMS frameworks and terminologies, which results in the lack of IMS presenting a universally accepted theoretical foundation. Even though there is significant interest in the application and impacts of ISO 56002, particularly its adoption and implementation, there is still insufficient examination of the standard, even so in previous versions from different countries, as seen in Cluster 4, Table 14. Despite the fact that

Table 13
Selection for the first category of Cluster 4.

Title	Year	Main Findings
Total Innovation Management competence and innovation performance in SMEs - An Empirical Study Based on an SME Survey in Zhejiang Province	2008	Innovation performance in SMEs.
Predicament and Countermeasure of Leveraging the Innovation Capability of SMEs in Fujian Province - A Study Based on the Theory of Total Innovation Management	2010	Innovation of small and medium-sized enterprises (SMEs) in Fujian province.
Improving the Innovation Management in Romanian SMEs	2010	Innovation management at the level of Romanian SMEs.
Customer-oriented innovation management in small and medium-sized businesses: Evaluation of an intervention in trade companies	2010	Survey on the characteristics of customer-oriented innovation management.
Exploring the Use of Innovation Performance Measurement to Build Innovation Capability in a Medical Device Company	2014	Innovation management and measurement theory are combined with empirical investigations.
Evaluating innovative processes in French firms: Methodological proposition for firm innovation capacity evaluation	2014	Innovation capacity of SMEs in Lorraine, France.
Innovation management capabilities in rural and urban knowledge intensive business services: empirical evidence	2015	Innovation management capabilities in effect at a knowledge-intensive business services (KIBS) firm.
Innovation Management and Innovation Potential of Croatian SMEs	2015	IMS on Croatian SMEs.
Innovation Management System of Ecuador	2015	IMS on Ecuador and implementation of a Triple Helix Model.
The Importance of the Dimensions of Innovation Management in Evaluating the Innovation Capability of Firms in the Machine Building Industry in Romania	2016	Evaluation of the innovation capability based on scores.
ERP-Based Innovation Management System for Engineered-to-Order Production	2016	Development of new products within an ETO enterprise.
New Approaches in Company Innovation Management and its Application in Slovakia	2017	SME innovation in Slovakia.
Innovation Capacity of City Administrations: A Best Practices Approach	2022	Innovation capability evaluation framework.
Effective Implementation of the Enterprise's Innovation Capacity	2023	Effectiveness of innovation activities in an enterprise.
Innovation in the Lubusz Region. Implementation of an Innovation Management System	2023	Activities and initiatives supporting innovation in the Lubusz Voivodeship.

Table 14
Selection of second category of Cluster 4.

Title	Year	Main Findings
Innovation management standards. A Comparative analysis	2011	Study of innovation in Spain and the UK.
Standardized innovation management systems: A case study of the Spanish Standard UNE 166002:2006	2011	Spanish manufacturing firm and UNE 166002:2006 Standard.
Certifying Innovation: A Proposal for a Standard with the Innovation Management System (IMS)	2012	QMS and IMS Certification.
Innovation Management (Une-Cen/Ts 16555–1:2013) Applied to Superior Education: Integration of Disruptive Technologies for the Teaching of Chemistry	2015	Study of the European Standard for Innovation Management.
The performance implications of the UNE 166.000 standardized innovation management system	2019	IMS and Spanish UNE 166.000 Standard.
Does adoption of ISO 56002–2019 and green innovation reporting enhance the firm's sustainable development goal performance? An emerging paradigm	2021	IMS and ISO 56002 on sustainable development.
AI-based audit of fuzzy front-end innovation using ISO56002	2021	FFE AI audit on IMS.
Improving Firm Innovation Capacity Through the Adoption of Standardized Innovation Management Systems: A Comparative Analysis of the ISO 56002:2019 with The Literature on Firm Innovation Capacity	2021	Innovation Capacity and ISO 56002:2019.
Self-Assessment Model based on the ISO 56002:2019 Standard for Evaluating Innovation Management Systems of Science and Technology Institutions	2022	ANP and IPA methods on ISO 56002 and IMS.
Effects of the UNE 166.002 standards on the incremental and radical product innovation and organizational performance	2023	New product development and UNE 166.002 Standards.

this research covers a significant amount of case studies, over-reliance from specific industries or regions limits the generalizability of findings, expressing the need for more longitudinal studies to understand the long-term impact of IMS. Most importantly, Small and medium-sized enterprises (SMEs) often struggle with resource constraints when implementing IMS, but this area is underexplored compared to larger organizations, with very little literature found on the subject, as observe in Tables 6 and 10, with only a handful of articles dedicated to this part.

In the first cluster, there is a trend toward integrating IMS with other organizational systems, such as Project Management, Knowledge Management and Product Development Management, in order to create an ideal innovation ecosystem. Successful IMS implementations prioritize employee engagement by encouraging a culture of innovation, by not only providing tools but also fostering a collaborative and supportive environment that encourages employees to contribute ideas. Besides the above, there is an increased recognition of the need for IMS solutions that are scalable and adaptable, as seen in the fourth cluster. Customized approaches that take into account the specific needs and resources of SMEs are gaining more attention, especially in the literature, with a diverse amount of case studies in different sectors, as observe in the second and third cluster.

In regards of the gaps in the literature, several opportunities present themselves for further research. Firstly, there is a lack of in-depth studies focusing on Small and Medium Enterprises (SMEs), leaving an interval in understanding how IMS can be effectively help the challenges and opportunities faced by these businesses, especially since the adoption of ISO 56002. Comparison between different industries is also limited, hindering insights into variations in IMS adoption and performance across different sectors. (Lerche, 2022)

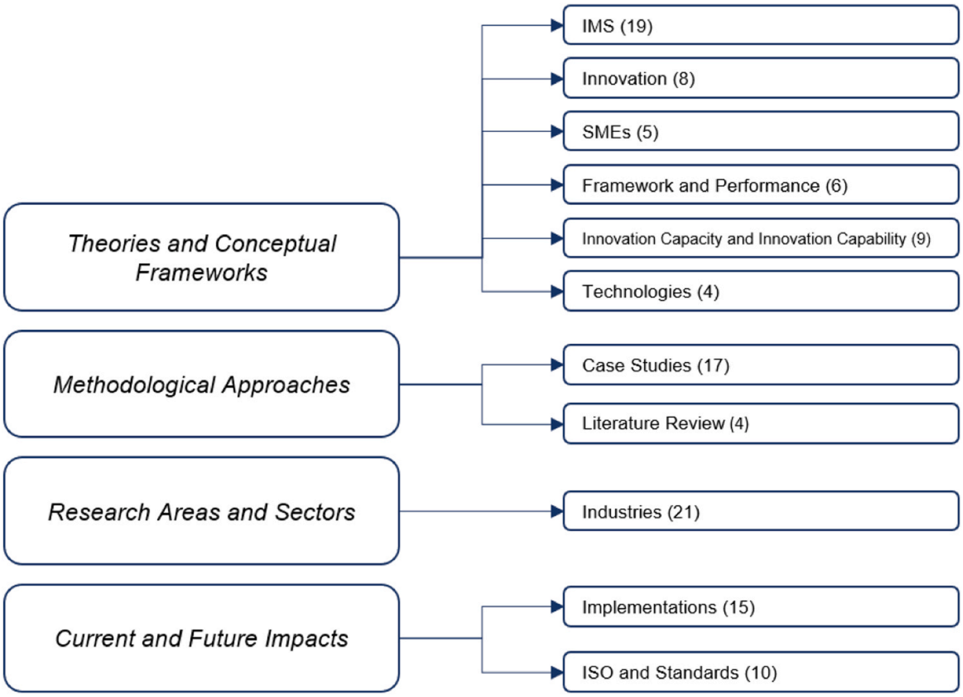

Fig. 8. Clusters and number of articles of each cluster.
Source: Author's own contribution.

also discussed the possibilities of applying ISO 56002 within organizations, which can also lead to new research that focuses more in the application of the standard. Additionally, studies focusing in the long-term effects of IMS implementation and its evolution over time are insufficient, considering that since 2019 there is a standard set to implement and maintain IMS, the family of ISO 56000. Regarding the above, the ISO 56002 strongly declares that the guidelines can apply to "all types of organizations, regardless of type, sector, or size. The focus is on established organizations, with the understanding that both temporary organizations and start-ups can also benefit" (ISO, 2021). Addressing these gaps could enhance our knowledge of IMS's adaptability and sustainability, particularly in the context of SMEs and diverse industry settings.

The proposed research agenda identifies open issues for each cluster that represent future lines of research. For the first cluster, *Theories and Conceptual Frameworks,* and based on the literature, the potential gaps, considering the theories and frameworks presented, include limited exploration of alternative or emerging theoretical perspectives in recent literature, lack of integration between theoretical frameworks and hindering a comprehensive understanding of IMS. Besides, the potential of IMS is limited by the research on the evolution of theories and frameworks over time, delaying the identification of current trends.

Regarding the *Methodological Approaches* Cluster, it is foreseen the limited use of mixed-methods approaches, potentially overlooking the nuanced aspects of IMS implementation. Also, insufficient attention is paid to the validation and reliability of measurement tools used in IMS research. The third cluster, *Research Areas and Sectors*, presents an underrepresentation of specific industries or sectors in the literature, leading to sector-specific knowledge gaps. This can lend to potential bias toward studying large organizations, neglecting the innovation challenges faced by small and medium-sized enterprises (SMEs). Additionally, it is noted in the literature insufficient research on the intersection of IMS with emerging industries or disruptive technologies. Finally, the fourth cluster, *Current and Future Impacts,* lacks of longitudinal studies tracking the long-term impact and sustainability of IMS implementations with a limited

Table 15
Future research agenda on IMS.

Cluster	Open Research Questions
Theories and Conceptual Frameworks	Considering the variety of theories adopted to investigate, How have these theories evolved over time, and what trends can be identified?
	Are there emerging or innovative theoretical approaches in recent literature on IMS?
Methodological Approaches	What methodological approaches can be used to assess the impact of IMS?
	Are there emerging trends in research methods for studying IMS?
Research Areas and Sectors	What specific research areas within IMS have received the most attention in the literature?
	Are there industry sectors that have been more extensively studied in the context of IMS?
Current and Future Impacts	What is the future impact of IMS on organizational innovation and performance?
	How are organizations adapting IMS to address current challenges and opportunities?

focus on the micro-level impacts within teams and individual employees. Most importantly, insufficient exploration of the impact of IMS on organizational culture and employee engagement is provided.

Table 15 shows the future research agenda on the subject of study. Addressing these questions in the literature and research will contribute to a more comprehensive understanding of IMS, inform future directions, and offer practical insights for organizations aiming to enhance their innovation management practices. Future research can strategically design studies to fill these gaps and contribute to the evolving field of IMS, hence the importance of highlight the proposed questions.

Emerging trends and challenges in IMS are identified through a combination of bibliometric analysis, content analysis, and qualitative synthesis of the literature. Bibliometric techniques, such as keyword co-occurrence and citation network analysis using VOSviewer, help detect evolving themes by highlighting frequently cited topics like digital transformation and open innovation. Content analysis of recent studies (especially post-2020) further reveals shifts in research focus, such as the increasing role of AI or agile methodologies in IMS. By examining how these themes have developed over time, our captures significant changes in scholarly perspectives.

The latest trends in Innovation Management Systems (IMS) include the integration of digital transformation, artificial intelligence, and data-driven decision-making to enhance innovation processes. There is also a growing focus on open innovation, where organizations collaborate with external partners, startups, and research institutions, like the works of (Garechana et al., 2017) and (Bogers et al., 2019). Sustainability and social innovation have gained prominence, with companies aligning their innovation strategies with environmental and social impact goals (; P. A. Khan et al., 2021). Additionally, agile methodologies and design thinking are being increasingly applied to innovation management to foster flexibility and rapid iteration (Giménez-Medina et al., 2023).

Key challenges include organizational resistance to change, as implementing IMS often requires cultural and structural shifts. Measuring innovation performance remains complex due to the intangible nature of innovation outcomes. Ensuring intellectual property protection in collaborative environments is another concern. Additionally, organizations face difficulties in balancing exploratory innovation (radical ideas) with exploitative innovation (incremental improvements). Intellectual Property (IP) Management refers to the strategies and practices organizations use to protect, leverage, and maximize the value of their intellectual assets (Peterková et al., 2020). These assets include patents, trademarks, copyrights, trade secrets, and proprietary knowledge.

Finally, our analysis primarily reflects how IMS has been conceptualized in scholarly work rather than a direct empirical comparison with industry practices because it relies on published academic literature as its primary data source. Systematic literature reviews synthesize existing studies, theories, and conceptual frameworks developed by researchers, often emphasizing definitions, models, and methodological approaches rather than real-world applications. Since the review does not include direct engagement with industry practitioners, case studies, or field data, its findings represent the academic discourse on IMS rather than firsthand insights from industry implementation.

Furthermore, the clustering process in bibliometric and content analysis groups articles based on their thematic connections within scholarly research. While some studies may discuss practical applications, their perspectives remain filtered through an academic lens, focusing on theoretical constructs, innovation strategies, and methodological advancements. This means the review highlights how IMS is structured and debated in research rather than assessing its effectiveness in organizations. To bridge this gap, future studies could incorporate empirical validation through surveys, interviews, or case studies to compare academic models with industry realities.

6. Conclusions

This paper aims to provide a comprehensive and systematic literature review on Innovation Management Systems (IMS). This research contributes to a thorough and integrated overview of the existing scientific literature. The analysis conducted throughout the review has successfully addressed key research questions, revealing a multifaceted understanding of IMS. The integration of various theoretical perspectives has enriched the discussion, highlighted the complexity of innovation management and offered a comprehensive framework for future exploration.

Concerning the development of the literature on Innovation Management System (Research Question 1), the number of articles being published saw a significant increase during the period 2003–2023, as shown in Fig. 3, with the distributions of papers per year. This growing attention, evidenced by the publication's temporal trend, is also confirmed by the evolving nature of the timing and number of citations (see Table 2 and Table 3). The trend of publications and the number of citations per year and the most cited authors confirm IMS as an evolving topic, which is gaining increasing attention from the scientific community. More importantly, this study also aims to propose the growth in the number of articles related to ISO 56002, as this standard is the outset of Innovation Management System, as its global acceptance makes it a valuable tool for organizations looking to implement consistent and standardized practices across all sectors and organizations. The standard incorporates best practices and provides detailed guidance on establishing and maintaining an effective IMS. Organizations can use ISO 56002 as a reference point for developing strategies, processes, and structures that align with recognized principles of innovation management. Developing robust metrics to evaluate the success of IMS in achieving innovation outcomes is a priority.

Regarding the emerging trends and challenges of the literature on IMS, and the understanding of IMS in time (Research Question 2 and 3), the analysis carried out identified four main research clusters: *Theories and Conceptual Frameworks*, *Methodological Approaches*, *Research Areas and Sectors* and *Current and Future Impacts.* With the examination of each cluster, it can be addressing that the gaps identified in the literature emphasize the need for future research to fill these voids. Conclude by proposing specific areas of exploration, such as the impact of IMS on organizational culture, the role of IMS in the context of disruptive technologies, and the innovation challenges faced by SMEs. Moreover, the analysis was carried out to identify the research agenda IMS, indicating potential themes for future exploration (Research Question 4). As presented in Table 15, a number of several open research questions have been proposed for each identified cluster within the literature. These suggestions offer valuable insights for researchers aiming to make additional contributions to this domain and for managers grappling with challenges associated with IMS.

A literature review on IMS faces several potential limitations. One major challenge is the limited scope of existing research, with gaps in geographic, industrial, and sectoral coverage. Variations in how IMS is defined or conceptualized across studies can lead to inconsistent interpretations and difficulty consolidating findings. Over-reliance on case studies, while offering depth, often limits generalizability, and a lack of longitudinal or quantitative studies makes it difficult to assess long-term impacts or establish causality. Another limitation is the emerging nature of IMS frameworks like ISO 56002, which restricts the availability of empirical data on their practical implementation and outcomes. The interdisciplinary nature of IMS, spanning management, technology, and strategy, introduces complexity in synthesizing diverse theories and methodologies. Furthermore, the rapid evolution of technology and societal trends can make older studies less relevant to current contexts. Addressing these gaps requires a broader scope of research, greater methodological rigor, and a focus on longitudinal and cross-industry studies to better understand IMS and its transformative potential.

Our study highlights that the conceptualization of IMS in academic literature is predominantly theoretical, focusing on frameworks, methodologies, and sector-specific discussions rather than direct empirical validation within industry settings. While this provides valuable insights into how IMS is structured and debated in research, it also underscores the need for further studies that bridge the gap between theory and practice. Future research could incorporate empirical methodologies such as case studies, expert interviews, or industry surveys to assess the practical implementation, effectiveness, and challenges of IMS in real-world organizations. By integrating both academic perspectives and industry experiences, future studies can offer a more comprehensive understanding of IMS and its role in driving innovation management success.

Finally, this research highlights the critical role of Innovation Management Systems (IMS) in fostering creativity, streamlining innovation processes, and aligning organizational goals with innovation strategies. By exploring key theoretical frameworks, methodological approaches, diverse research areas, and future-oriented impacts, this study provides a comprehensive understanding of the existing body of knowledge. However, the review also identifies notable gaps, such as inconsistencies in definitions, limited empirical studies, and uneven sectoral coverage, which present significant opportunities for further research. As IMS continues to evolve, addressing these gaps and incorporating emerging trends will be essential for advancing both academic insights and practical applications, ensuring organizations can fully leverage IMS to achieve sustainable innovation.

CRediT authorship contribution statement

Karstegl Villalobos Natalia Francisca: Writing – original draft, Project administration, Investigation, Conceptualization. **Frédérique Mayer:** Validation, Project administration, Formal analysis. **Manon Enjolras:** Writing – review & editing, Visualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

Data availability

No data was used for the research described in the article.

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